

SIORALIK, Greenland

WMO: 042420

Lat: 65.013N

Lon: 52.529W

Elev: 14

StdP: 101.16

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.9	-16.8	-22.9	0.5	-18.3	-21.0	0.6	-16.1	21.2	-1.1	18.3	-4.9	6.9	60	0.973

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.8	11.6	8.9	10.5	8.4	9.6	7.9	9.6	10.9	8.8	9.9	8.1	9.2	4.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.7	7.0	10.1	8.0	6.7	9.3	7.3	6.4	8.7	28.4	11.1	26.7	10.1	25.2	9.3	13.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	14.2	12.4	DB	-19.6	15.7	4.1	2.6	-22.6	17.6	-25.0	19.1	-27.3	20.6	-30.3	22.4	(4)
(5)			WB	-20.2	11.2	3.6	1.5	-22.8	12.3	-25.0	13.1	-27.0	13.9	-29.7	15.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-6.4	-7.7	-7.4	-2.9	1.1	3.9	6.4	6.8	4.9	1.0	-2.2	-4.5	(6)
(7)		DBStd	6.22	4.74	5.29	5.44	3.69	2.33	2.25	2.01	1.77	1.99	2.84	3.75	4.12	(7)
(8)		HDD10.0	3852	508	494	539	387	274	183	113	102	154	280	366	450	(8)
(9)		HDD18.3	6889	766	728	797	637	533	433	370	359	404	538	616	708	(9)
(10)		CDD10.0	4	0	0	0	0	0	1	2	2	1	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.2	7.6	7.4	7.1	6.3	5.2	4.8	4.6	4.8	6.0	6.2	7.2	7.1	(14)
(15)	Precipitation	PrecAvg	590	57	36	35	36	32	36	47	65	66	56	64	61	(15)
(16)		PrecMax	822	157	86	92	81	58	75	91	109	142	117	134	128	(16)
(17)		PrecMin	392	5	14	3	13	8	12	18	23	19	16	24	26	(17)
(18)		PrecStd	106	35	17	21	18	14	15	21	22	26	22	27	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	4.9	4.5	6.0	8.4	12.0	13.8	13.5	11.8	9.2	8.0	5.8	(19)
(20)			MCWB	2.9	2.0	2.4	2.7	5.5	8.3	9.9	10.6	7.8	5.5	5.7	3.5	(20)
(21)		2%	DB	2.9	3.0	2.7	3.9	6.5	9.8	11.4	11.5	10.0	6.7	5.8	3.6	(21)
(22)			MCWB	0.9	1.1	0.9	1.9	4.5	7.1	8.8	9.4	8.0	4.4	3.8	2.1	(22)
(23)		5%	DB	1.6	1.1	1.3	2.8	5.2	8.2	10.3	10.4	8.6	5.5	4.2	2.4	(23)
(24)			MCWB	-0.1	-0.5	-0.1	1.2	3.7	6.6	8.4	8.9	6.8	3.6	2.4	1.0	(24)
(25)		10%	DB	0.0	-0.4	0.1	1.8	4.1	7.0	9.4	9.4	7.4	4.3	2.6	0.9	(25)
(26)			MCWB	-1.4	-1.6	-1.1	0.7	3.0	5.7	7.9	8.2	5.8	2.8	1.1	-0.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	2.9	2.7	3.5	6.2	8.7	10.4	11.1	9.3	6.5	6.2	3.9	(27)
(28)			MCDB	4.3	4.3	4.1	4.9	7.6	11.1	12.5	12.8	10.3	8.3	7.6	5.3	(28)
(29)		2%	WB	1.5	1.2	1.1	2.4	4.8	7.5	9.3	9.9	8.2	4.9	4.4	2.5	(29)
(30)			MCDB	2.5	2.7	2.1	3.5	5.9	9.2	10.7	11.0	9.5	6.4	5.6	3.4	(30)
(31)		5%	WB	0.0	-0.4	0.1	1.6	3.9	6.6	8.6	9.2	7.0	3.8	2.6	1.2	(31)
(32)			MCDB	1.2	1.0	1.2	2.5	4.9	8.1	9.9	10.1	8.3	4.9	3.8	2.1	(32)
(33)		10%	WB	-1.3	-1.6	-1.0	0.8	3.1	5.8	8.0	8.3	6.1	3.1	1.3	-0.2	(33)
(34)			MCDB	0.1	-0.5	0.1	1.6	4.0	7.0	9.2	9.2	7.0	4.2	2.5	0.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.3	3.5	3.6	3.3	3.0	3.1	3.2	2.8	2.5	2.4	2.6	2.9	(35)
(36)			MCDBR	4.9	5.6	4.1	4.2	5.1	5.5	5.0	4.6	4.0	3.7	4.0	3.8	(36)
(37)			MCWBR	4.1	4.6	3.4	3.2	3.3	3.4	3.1	3.0	2.5	2.8	3.2	3.3	(37)
(38)		5% WB	MCDBR	5.1	5.3	4.1	3.9	4.8	5.0	4.5	3.9	3.4	3.4	3.7	3.5	(38)
(39)			MCWBR	4.6	4.8	3.5	3.2	3.4	3.4	3.1	2.8	2.4	2.8	3.4	3.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	N/A	0.244	0.276	0.324	0.338	0.346	0.346	0.348	0.308	0.265	0.353	0.350	(40)	
(41)		taud	N/A	2.280	2.355	2.286	2.350	2.441	2.483	2.484	2.525	2.351	2.816	2.891	(41)	
(42)		Ebn at Noon	N/A	685	820	845	868	868	858	820	784	641	339	196	(42)	
(43)		Edh at Noon	N/A	49	73	101	104	98	92	83	64	46	21	13	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.09	0.55	1.80	3.51	4.92	5.41	4.95	3.60	2.19	0.95	0.19	0.03	(44)	
(45)		RadStd	0.01	0.07	0.20	0.32	0.34	0.49	0.53	0.28	0.27	0.09	0.04	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (2 neighbors)	+0.49	+1.63	N/A	+1.12	N/A	N/A	N/A	N/A	+11	N/A	(47)	

Nomenclature: See separate page